

SUBMITTAL INDEX

For Approval · JANU-SUB-011 · Version V01

Project:	Janu Resort - Turks & Caicos Islands	Spec Section:	GP1R-E4 & GP1R-E5
Submittal No.:	JANU-SUB-011 · Rev V01	Submitted To:	BCQS Ltd - Contract Administrator
Date:	18 September 2026	Type:	Product Data
Contractor:	Waratah Construction TCI Ltd	Re:	Electrical distribution equipment - Mockup Room 1

Contents of This Submittal

This submittal covers the electrical distribution equipment for GP1 Mockup Room 1: load centers, meter socket, surge protection device, and molded-case / plug-in circuit breakers, as detailed on the attached ABB manufacturer data. Source: Utopia Electrical Ltd, Submittal No. 011 Rev. B (9/15/2026). Items 1 to 3 are substitutions and item 4 is a deviation (NEMA 4 vs specified NEMA 3R); these require engineer confirmation before release for fabrication or installation. Manufacturer data follows on pages 2 to 11; each item's page location is given in the Pg. column and repeated as a comment on the page.

Item Index

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A - ELECTRICAL DISTRIBUTION EQUIPMENT · ABB (SOURCE: UTOPIA ELECTRICAL SUB-086)				
1	Load Center, ReliaHome PCC4022R (Outdoor 3R, 225A, 40/80)	ABB PCC4022R	175A main kit THQMV175F	2
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3	Meter Socket, PAMS3022TRNUS (Outdoor 3R, 225A, 30/60)	ABB PAMS3022TRNUS	32-hole ground bar TKG32	4
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11	MCCB, 2P 30A (Qty 2)	ABB THQL2130	-	11

FOR:

☐ Approved ☐ Approved as Noted ☐ Revise and Resubmit ☐ Rejected

Reviewed By: _____

Date: _____

Project:	Janu Resort - GP1 - Mockup Room	Spec Section:	GP1R-E4 & GP1R-E5
Submittal No.:	JANU-SUB-011	Submitted To:	Waratah Ltd
Date:	9/15/2026	Re:	Electrical Distribution Equipment
Contractor:	Utopia Electrical Ltd		

Contents of This Submittal

This submittal covers the load centers, meter socket, surge protection device, and molded-case / plug-in circuit breakers specified for Mockup Room 1, as detailed on the attached manufacturer data (2 pages). Substitutions from the original specification are flagged in red beside the affected line items and require engineer confirmation before release for fabrication or installation.

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| 1 | Load Center, ReliaHome, PCC4022R | <i>see note</i> |
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| 6 | MCCB, 120/240V QL, THQL2180 | |
| 7 | MCCB, 120/240V QL, THQL2140 (qty 2) | |
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| 10 | MCCB, 120/240V QL, THQL2120 | |
| 11 | MCCB, 120/240V QL, THQL2130 (qty 2) | |

FOR:

☐ Approved ☐ Approved as Noted ☐ Revise and Resubmit ☐ Rejected

Reviewed By: _____ Date: _____

Item #	Qty	Description	Catalog No	
1	1	Load Center, ReliaHome, PCC4022R		<i>Offered in place of a 42-space recessed NEMA 3R panel: ABB's nearest size is 40 spaces, and the NEMA 3R unit is surface mount only.</i>
	1	Outdoor Type 3R Enclosure 225 Amps, Convertible, 1 Phase, 3 Wire, 120/240, 120/208V ac, Space: 40, Circuit: 80 22 kA Minimum Interrupt Rating, Fully Rated, #1 - 300 Cu/Al Main Wire Size (AWG/kcmil) Cu-Al Factory Installed Equipment Ground Kit: TGK42,TNLK20 Dimension: 39.29 in (H) 14.88 in (W) 5.29 in (D) Modifications	<u>PCC4022R</u>	
	1	175A RELIAHOME P-SERIES LOAD CTR MAIN BKR KIT 22K, THQMV175F	<u>THQMV175F</u>	
2	1	Load Center, ReliaHome, PCC1612		<i>Sized to the 1'-4" x 1'-9" panel opening shown on drawing JN-GP1R-A11-07A, Section 01. 125A is the largest bus ABB offers in an enclosure that fits that opening; a 150A bus is 26 in high. Panel capacity is 32 circuits. Mounted in the load center above as the 80A 2-pole main, per Section 01. ABB's amp-rated main kits start at 100A, so the 80A main is supplied as a base kit plus this breaker.</i>
	1	NEMA 1 Indoor Enclosure, Combination 125A Copper Bus, Convertible Main, 120/240V ac, 1 Phase, 3 Wire, 22 kAIC 16 Spaces, 32 Circuits Surface/Flush Cover Equipment Ground Kit Factory Installed: TGK12, TNLK20 Dimension: 20.38 in (H) 15.36 in (W) 3.79 in (D) Modifications	<u>PCC1612</u>	
	1	80A 2-Pole Main Breaker, 120/240 Vac, 22 kAIC	<u>THHQL2180</u>	
	1	ReliaHome Breaker Base Kit, 22 kAIC Converts to Main-Breaker Panel	<u>TQMH000F</u>	
3	1	Load Center, Meter Socket, PAMS3022TRNUS		<i>Offered with aluminum bus and a 225A main in place of the specified copper bus / 175A main - ABB's meter combo is not offered in that combination.</i>
	1	Outdoor Type 3R Enclosure, Surface 225 Amps, Main Circuit Breaker, 1 Phase, 3 Wire, 120/240, 120/208V ac, Space: 30, Circuit: 60 22 kA Minimum Interrupt Rating, Fully Rated, 1-300 Main Wire Size (AWG/kcmil) Cu-Al Dimension: 53.05 in (H) 14.93 in (W) 5.81 in (D) Modifications	<u>PAMS3022TRNUS</u>	
	1	RELIAHOME 32-HOLE EQUIP GROUND BAR, TGK32	<u>TGK32</u>	

Item #	Qty	Description	Catalog No
4	1	Surge Protection Device, Wall Mount, RG Series	
	1	UL SPD Type : Type 1 Surge Current Rating : 80 kA per mode/160 kA per phase Voltage : 120/240 Fault Withstand : 200 kAIC Enclosure : NEMA 4 W/O Disconnect Integral Disconnect : No Pre-Wired/Pigtail Leads : No Features : Rotatable Color View LCD display Dry Contacts Service Life indicator (LED) Phase loss indication N-G Voltage Detection > 20V detected Surge, outage, And TOV counter Phase loss indication Audible Alarm	RGWMSP120S08T14 <i>NEMA 4 enclosure exceeds the specified NEMA 3R. MCOV to be confirmed against the specified 140-150V on the manufacturer submittal.</i>
5	33	Circuit Breaker, Arc Fault Plug In, THQL1120P.	THQL1120PAF2
		1P 20AMP 10KAIC PON ARC FAULT	
6	1	MCCB, 120/240V QL, THQL2180	THQL2180
		THQL 2 POLE 120/240V 10K IC 80 AMP	
7	2	MCCB, 120/240V QL, THQL2140	THQL2140
		THQL 2POLE 120/240V 10K IC 40 AMPS	
9	1	Circuit Breaker, Ground Fault Plug In, THQL21	THQL2160GFT2
		THQL 2P 120/240V 60A GFT2	
10	1	MCCB, 120/240V QL, THQL2120	THQL2120
		THQL 2 POLE 120/240V 10K IC 20 AMP	
11	2	MCCB, 120/240V QL, THQL2130	THQL2130
		THQL 2 POLE 120/240V 10K IC 30 AMP	

Load Center, ReliaHome PCC4022R

Catalog No.	PCC4022R
Enclosure	Outdoor Type 3R
Rating	225 A convertible, 1-ph 3-W, 120/240
Spaces / circuits	40 / 80
Interrupt	22 kAIC, fully rated
Main wire	#1-300 Cu/Al
Ground kit	TGK42, TNLK20
Dimensions	39.29 x 14.88 x 5.29 in
Modification	175A ReliaHome P-Series main breaker kit - THQMV175F

Substitution / Deviation - engineer confirmation required

Offered in place of a 42-space recessed NEMA 3R panel: ABB's nearest size is 40 spaces, and the NEMA 3R unit is surface-mount only. Requires engineer confirmation.



Representative Image

Catalog No. PCC4022
Description: LD CTR CONV 225A 40/80 CU N1 GR PON UL
UPC No 783164692730
Home > Load Center & Meter Combos > Load Centers

ABB ReliaHome makes homes reliable, with ease. We designed our ReliaHome load centers to be installed as fast as possible - and reliable so your customers can leave a job with confidence. They work seamlessly with all our residential circuit breakers, optimizing your stock.

Descriptors

Category	Load Centers
GO Schedule	RV
Grouping	Single Phase - USA

Specifications

# Branch Circuit 250A	40
1" Space	40
1/2" Space	0
Amperage	225 A
Circuit Breaker Style	Plug-On Neutral
Door	Yes
Enclosure	Indoor
Enclosure Type	Indoor, Combination
Factory Installed Ground Kit	YES
Feed	Either
Input/Output Voltage	120/240V ac
Interrupt Rating	22kAIC
Interrupting Capacity Rating	22kAIC
Load Center Type	P-Series+ (PON)
Meter Socket Options: 200 Amp Main Breaker Installed	No
Meter Socket Options: High Amp Mode	None
NEMA Rating	NEMA 1
No of Circuits	80
Phase	1 Phase
Service & Voltage	1 Phase, 3 Wire, 120/240V ac
Wires	3 Wire

Classifications

UL/CUL Listing	UL
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by ABB

RESIDENTIAL MAIN CIRCUIT BREAKERS



100 to 225A. THQMV Main breakers are configured to allow top or bottom modification and can be added to ReliaHome™ P-Series, P-Series+, T-Series, PowerMark™ Gold and PowerMark Pro convertible load centers.

FEATURES AND BENEFITS

- Every breaker exceeds UL, NEMA, ANSI, CSA requirements
- 65° C/75° C Conductor Rating
- Handle trips to center position, giving a clear indication of a tripped breaker
- Handle requires reasonable effort to operate
- Handle's vertical orientation allows it to be mounted for top or bottom feed applications
- 22KAIC interrupting capacity

APPLICATIONS

- 200 amp ReliaHome P-Series and P-Series+
- 200 amp legacy PowerMark Gold Pro convertible load centers
- For 125 amp P-Series and P-Series+ convertible load centers. use TQMH000F and desired plug-on breaker (THHQL21XX)
- For 125 amp legacy PowerMark Gold Pro convertible load centers. Use TQMH000 and desired plug on breaker (THHQL21XX)

Load Center, ReliaHome PCC1612

Catalog No.	PCC1612
Enclosure	NEMA 1 Indoor, Combination
Bus / main	125A copper bus, convertible main, 120/240, 1-ph 3-W, 22 kAIC
Spaces / circuits	16 / 32
Cover	Surface / flush
Ground kit	TGK12, TNLK20
Dimensions	20.38 x 15.36 x 3.79 in
Modifications	80A 2-pole main THHQL2180 + ReliaHome breaker base kit TQMH000F

Substitution / Deviation - engineer confirmation required

Sized to the 1'-4" x 1'-9" panel opening on drawing JN-GP1R-A11-07A, Section 01. 125A is the largest bus ABB offers in an enclosure fitting that opening (a 150A bus is 26 in high); capacity 32 circuits. The 80A 2-pole main is supplied as a base kit (TQMH000F) plus breaker (THHQL2180). Requires engineer confirmation.



Representative Image

Catalog No. PCC1612
Description: LD CTR CONV 125A 16/32 CU N1 GR PON UL
UPC No 783164692556
Home > Load Center & Meter Combos > Load Centers

ABB ReliaHome makes homes reliable, with ease. We designed our ReliaHome load centers to be installed as fast as possible - and reliable so your customers can leave a job with confidence. They work seamlessly with all our residential circuit breakers, optimizing your stock.

Descriptors

Category	Load Centers
GO Schedule	RV
Grouping	Single Phase - USA

Specifications

# Branch Circuit 250A	16
1" Space	16
1/2" Space	0
Amperage	125 A
Circuit Breaker Style	Plug-On Neutral
Door	Yes
Enclosure	Indoor
Enclosure Type	Indoor, Combination
Factory Installed Ground Kit	YES
Feed	Either
Input/Output Voltage	120/240V ac
Interrupt Rating	22kAIC
Interrupting Capacity Rating	22kAIC
Load Center Type	P-Series+ (PON)
Meter Socket Options: 200 Amp Main Breaker Installed	No
Meter Socket Options: High Amp Mode	None
NEMA Rating	NEMA 1
No of Circuits	32
Phase	1 Phase
Service & Voltage	1 Phase, 3 Wire, 120/240V ac
Wires	3 Wire

Classifications

UL/CUL Listing	UL
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by ABB



Catalog No. **THHQL2180**

Description: THHQL 2P 120/240V 22K IC 80 AMP

UPC No 783164092059

Home > Circuit Breakers > Residential Circuit Breakers > Q Line

Q line circuit breakers are one-inch wide per pole, compact, thermal-magnetic devices designed for residential and commercial applications. The QL breakers are plug-in versions of the Q Line used for connection to load centers and lighting panels. All Q Line circuit breakers feature Quick-make / Quick-break mechanisms, common trip bars, and easy to spot trip indication to ensure safety and reliability. Q Line breakers are available in 1, 2, and 3 pole versions, can be ordered with auxiliary contact and shunt trip accessories, and can be ordered for use in HID applications. 2180 Amps 80 A

Descriptors

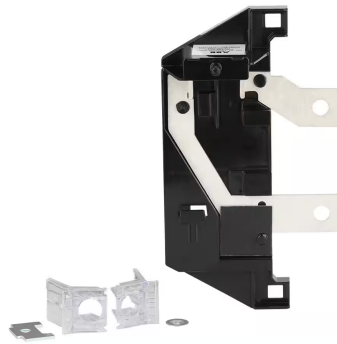
Category	Q Line
Product Line	Q-Line (Plug-In)
GO Schedule	R5

Specifications

Interrupting Capacity Rating	22 kAIC
Voltage	120/240 V
Trip Style	Non-Interchangeable
Frame Type	Q-Line
Amperage	80 A
System Voltage	120 Vac 120/240 Vac
Poles	2
Trip Function	LI
Continuous Current Rated	Standard
120 Vac Interrupting Rating	22 KAIC
120/240 Vac Interrupting Rating	22 KAIC
Suitable for Reverse Feed	Yes
Long Time	Fixed
Instantaneous	Fixed
Protective Relays	No
Current Metering	No
Special Markings	HACR
GSA Compliance	No

Classifications

UL File #	E11592
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PRODUCT-DETAILS

TQMH000F**MAIN BKR BASE ONLY KIT 125A MAX****General Information**

Global Commercial Alias:	TQMH000F
Extended Product Type:	TQMH000F
Product ID:	1TQQ12000X0301
Catalog Description:	MAIN BKR BASE ONLY KIT 125A MAX
Long Description:	Q-line plug-in circuit interrupters are one-inch wide per pole, compact, thermal-magnetic devices designed for residential and commercial applications in ReliaHome™ load centers. These breakers feature quick-make / quick-break mechanisms and easy to spot trip indication to ensure safety and reliability. Thermal magnetic circuit interrupters provide protection against overloads and short circuits.
Display Name:	TQMH000F

Material Compliance

RoHS Information:	No declaration needed
REACH Information:	No declaration needed
Conflict Minerals Reporting Template (CMRT):	9AKK108468A3363
WEEE Category:	Product Not in WEEE Scope

Certificates and Declarations

Declaration of Conformity - CE:	No declaration needed
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Installation

Instructions and Manuals:	No document needed
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Popular Downloads

Data Sheet, Technical Information:	No document needed
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External Classifications and Standards

ETIM 9:	EC000228 - Power circuit-breaker for trafo/generator/installation protection
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ETIM 10:	EC000228 - Power circuit-breaker for trafo/generator/installation protection
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Categories

Products > Low Voltage Products and Systems > NEMA Residential and Commercial Portfolio > Residential Breakers > MAIN CB

Load Center, Meter Socket PAMS3022TRNUS

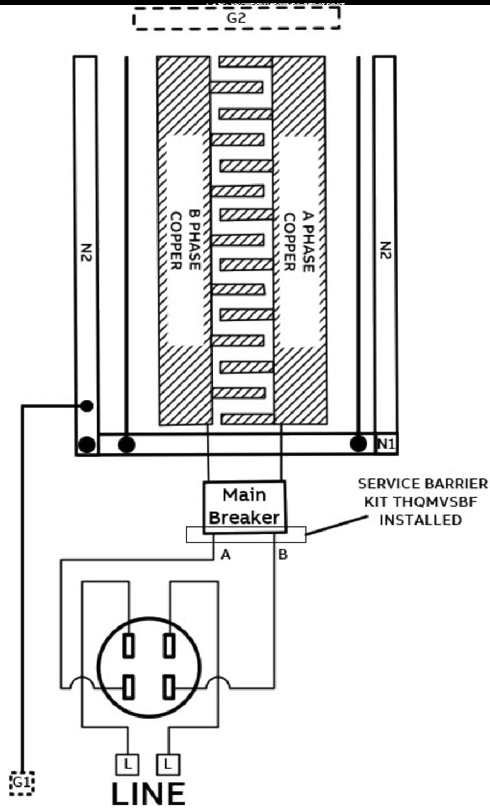
Catalog No.	PAMS3022TRNUS
Enclosure	Outdoor Type 3R, Surface
Rating	225 A main breaker, 1-ph 3-W, 120/240
Spaces / circuits	30 / 60
Interrupt	22 kAIC, fully rated
Main wire	1-300 Cu-Al
Dimensions	53.05 x 14.93 x 5.81 in
Modification	ReliaHome 32-hole equipment ground bar - TGK32

Substitution / Deviation - engineer confirmation required

Offered with aluminum bus and a 225A main in place of the specified copper bus / 175A main - ABB's meter combo is not offered in that combination. Requires engineer confirmation.

ABB**ReliaHome™****Meter Combo**

ENCLOSED
PANELBOARD
ISSUE NO. C-3024

**SUITABLE ONLY FOR USE AS SERVICE EQUIPMENT.**

MAXIMUM CONTINUOUS LOAD ON BRANCH CIRCUIT NOT TO EXCEED 80 PERCENT OF THEIR RATING.

ADDITIONAL/REPLACEMENT DEVICE SHALL BE OF SAME DESIGNATION AND OF EQUAL OR GREATER INTERRUPTING RATING

100 AMPS MAX BRANCH BREAKER
200 AMPS MAXIMUM STAB RATING

FOR ADDITIONAL SERIES RATING INFORMATION REFER TO DET008

BUILDING PAPER OR FLASHING IS TO BE OVER THE FRONT OF FLANGE WHEN SEMI-FLUSH DEVICE IS INSTALLED

THIS DEVICE MEETS E.U.S.E.R.C. SPECIFICATIONS

**DANGER**

HAZARD OF ELECTRICAL SHOCK OR BURN.
TURN OFF POWER BEFORE WORKING INSIDE
THIS EQUIPMENT.

RAINPROOF CONDUIT HUBS

WHEN CABINET HAS PROVISIONS FOR A CONDUIT HUB, SELECT FROM THE CHART BELOW.

DIAMETER (INCHES)	3/4	1	1 1/4	1 1/2	2	2 1/2	3	CAP	11/2	2
RAINTIGHT CAT.NO.	TC75	TC100	TC125	TC150	TC200	TC250	TC300	TCCP	TCCP15	TCCP20

THE SHORT CIRCUIT RATING OF THIS PANELBOARD IS EQUAL TO 22,000 RMS SYMMETRICAL AMPERES AT 240 V, SINGLE PHASE. WATT-HOUR METER NOT INCLUDED IN SHORT-CIRCUIT RATING.

METER COMBO

CAT NO. SURFACE TYPE : **PAMS3022TRNUS** MOD 1

225 AMPS MAX. 120/240V AC

3 WIRES, 1 PHASE

TYPE 3R ENCLOSURE

METER SOCKET RATING: 225 AMPS (200 AMPS CONTINUOUS)

MAIN BREAKER: THQMV225FB

USE ONLY ABB TYPE BREAKERS :

THQL, THHQL, THQLT, THQLQ

OVER CURRENT PROTECTION

IF BREAKER TRIPS, HANDLE WILL BE IN INTERMEDIATE POSITION. TO RESTORE SERVICE;

1. MOVE HANDLE TO "OFF" POSITION

2. THEN TO "ON" POSITION

USE COPPER OR ALUMINIUM

USE 60 °C OR 75 °C AMPACITY SIZED WIRE ON LINE NEUTRAL TERMINALS. SEE CIRCUIT BREAKERS FOR THEIR TEMPERATURE RATING.

WIRE RANGE TABLE

CONNECTOR	COPPER		ALUMINIUM	
	SOLID	STRD	SOLID	STRD
NEUTRAL – N1	-	1 – 300	-	1 – 300
NEUTRAL – N2	14 – 10	14 – 4	12 – 10	12 – 4
LINE – L	-	1 – 300	-	2/0 – 300
GROUND – G1	-	1 – 300	-	1 – 300
GROUND – G2	14 – 10	14 – 4	12 – 10	12 – 4

LUG TIGHTENING TORQUE (LB-IN)

SLOTTED SCREW		INTERNAL HEX SCREW	
AWG WIRE	TORQUE (LB-IN)	HEX SIZE	TORQUE (LB-IN)
14 - 10	32 - 35	3/16 ACROSS FLATS	108 - 120
8	36 - 40	1/4 ACROSS FLATS	180 - 200
6 - 4	41 - 45	5/16 ACROSS FLATS	248 - 275
3 - 2/0	45 - 50	-----	-----

ACCESSORIES

EQUIPMENT GROUND KITS	TGK32 , TGK42
FRONT FILLER PLATES	1" WIDTH - TQLFP1

WARNING: TO AVOID ACCIDENTAL CONTACT WITH LIVE PARTS, CLOSE ALL OPENINGS WITH BREAKERS OR FILLER PLATES. THE CIRCUIT BREAKERS IN THIS BOX HAVE BEEN SELECTED TO PROTECT THE WIRING TO AVOID DAMAGE AND LOSS OF PROTECTION, DO NOT REPLACE WITH HIGHER AMPERE RATED BREAKERS. FOR REPAIRS OR ALTERATIONS, CALL AN ELECTRICAL CONTRACTOR OR ELECTRICIAN.

BUSHING PLATE

INSTRUCTION SHEET
1TQC183000Z2401



ABB INC., GREGSON DRIVE,
CARY, INC, U.S.A
SALXXXX@
COUNTRY OF ORIGIN : MEXICO
10096008P37 REV. A



Catalog No. **TKG32**

Description: RH 32-HOLE GROUND

UPC No 783164158489

Home > Load Center & Meter Combos > Accessories/Parts

The **TKG32** is a 32 hole ground bar kit. For 6-12 circuit indoor load centers, utilize PowerMark Plus style neutrals and grounds. Add the TKGIS for an isolated ground and mount to ground kit.

Descriptors	
Category	Accessories/Parts
GO Schedule	RA
Grouping	Accessories/Parts

Surge Protection Device, Wall Mount, RG Series - RGWMSP120S08T14

Catalog No.	RGWMSP120S08T14
UL SPD Type	Type 1
Surge current	80 kA/mode; 160 kA/phase
Voltage	120/240
Fault withstand	200 kAIC
Enclosure	NEMA 4 (without disconnect)
Features	Colour LCD, dry contacts, service-life LED, phase-loss, counters, audible alarm

Substitution / Deviation - engineer confirmation required

Deviation: NEMA 4 enclosure exceeds the specified NEMA 3R. MCOV to be confirmed against the specified 140-150V. Requires engineer confirmation.

Model **RGWMSP** for ABB Wall Mounted Surge Protection

ReliaGear® Wall Mounted Surge Protective Devices (SPD)

Differentiating Technology

The ReliaGear® **RGWMSP** Series offers robust surge protection with an emphasis on performance, safe operation, and user-friendly diagnostics.



Standards

- UL 1449 5th Ed. – Listed, Open Type—Mil-Std-220C
- UL 1283 Listed EMI/RFI Filter—ANSI/NFPA 70 (Art. 242, Formerly 285)
- UL 96A – Meets SPD criteria—ANSI/IEEE C62.41 & .45
- cUL Listed to UL 1449 & CSA C22.2—ANSI/IEEE C62.62 & C62.72
- IEC 61643-11

ReliaGear® **RGWMSP** Series SPDs are engineered with smart thermal disconnecting technology that safely removes overstressed components before harmful external effects arise. This preventive approach enhances safety and avoids nuisance from OCPD tripping or thermal events.

The core suppression system includes proprietary Metal Oxide Varistors (MOVs) for surge energy diversion and metal film capacitors for filtering low-level transient voltages and electrical noise.

To support ongoing monitoring, the product includes an **LED status system**:

- **Red LED** indicates complete depletion and triggers replacement.
- **Orange LED** signals degraded but operational protection—giving the user a window to replace the SPD before total failure.

Additionally, a **rotatable LCD display** provides real-time feedback on key system parameters, including:

- Surge event counts
- TOV incidents
- Power outages
- Neutral-to-ground voltage over 20 V
- Remaining MOV life

This combination of protection, diagnostics, and replaceability makes the **RGWMSP** Series highly suitable for demanding power distribution applications.





Electrical Specifications

Surge Current Rating – (there is no standardized test from UL, IEEE or IEC).

Values in the table below are **per mode**.

Model	L-N	L-G	N-G	L-L
RGWMSPXXX08XX	80 kA	80 kA	80 kA	80 kA
RGWMSPXXX15XX	150 kA	150 kA	150 kA	150 kA
RGWMSPXXX22XX	225 kA	225 kA	225 kA	225 kA
RGWMSPXXX30XX	300 kA	300 kA	300 kA	300 kA

*225 & 300kA only available in NEMA 4 and 4X enclosures

Nominal Discharge Current	(In) 20kA all Models (UL 1449 Sec. 41.7; UL 96A Sec. 13; IEC 61643 Class II.)
Operating Frequency	50 Hz – 60 Hz
Modes of Protection	L-N, L-G, N-G, L-L
Short Circuit Current Rating	(SCCR) – 200 kA for all models
VPR & MCOV	See Table Below

MCOV Table

Model	Voltage	L-L	L-N	L-G	N-G	H-L	H-N	H-G
RGWMSP120SXXXX	120S	300	150	150	150			
RGWMSP120YXXXX	120Y	300	150	150	150			
RGWMSP240DXXXX	240D	540		270				
RGWMSP240HXXXX	240H	300	150	150	150	420	270	270
RGWMSP277YXXXX	277Y	640	320	320	320			
RGWMSP347YXXXX	347Y	840	420	420	420			
RGWMSP480DXXXX	480D	1100		550				
RGWMSP600DXXXX	600D	1380		690				

VPR Table

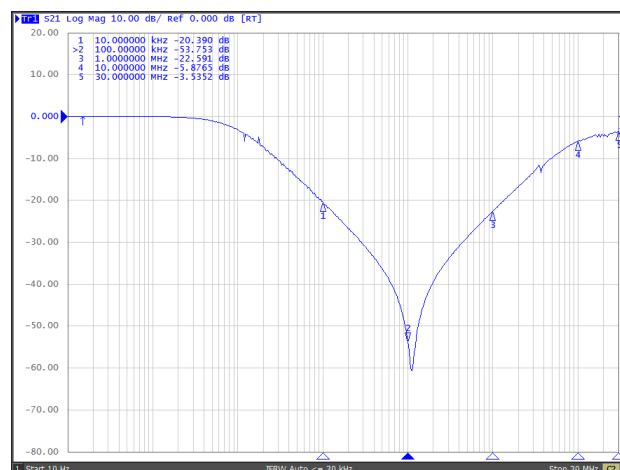
Model	Voltage	L-L	L-N	L-G	N-G	H-L	H-N	H-G
RGWMSP120SXXXX	120S	1200	700	700	700			
RGWMSP120YXXXX	120Y	1000	700	700	700			
RGWMSP240DXXXX	240D	1500		1000				
RGWMSP240HXXXX	240H	1200	700	800	700	1500	900	900
RGWMSP277YXXXX	277Y	2000	1200	1200	1200			
RGWMSP347YXXXX	347Y	2500	1500	1500	1500			
RGWMSP480DXXXX	480D	3000		1800				
RGWMSP600DXXXX	600D	4000		2000				

EMI/RFI Noise Attenuation (UL 1283 listed filter; UL Type 2 SPD Only).

Mil-Std-220C – US Dept of Defense Std for Method of Insertion Loss Measurement.

In a typical electrical system, frequencies above 30 MHz are radiated as opposed to conducted electrical noise; any measurement values above 30 MHz are considered irrelevant and should not be considered for specifying SPD filtering capability.

The overall frequency attenuation for all ReliaGear® SPD products is -3dB between 1 kHz and 30 MHz with the maximum attenuation at -61dB at 100.2 kHz.



Electrical Specifications

Fusing

All metal oxide varistors (MOVs) are thermally fused, providing event-free disconnection of the MOV from the electrical system upon the component's end of life. SPD status indication circuits (LEDs, Dry (volt-free) relay contacts, and Audible Alarms depending on features) give local and, if installed, remote status indication.

OCPD (Disconnect Switch)

The **RGWMSP** series is integrated with a Surge Rated Disconnect Switch.

SPD Status Monitoring

- Tri-Color LED indicators per phase (See Truth Table in Troubleshooting section).
 - Green LED per phase showing power applied and complete operational status of the SPD.
 - Red LED (one) when lit, the SPD componentry is compromised as one or more of the internal disconnectors have operated indicating that the SPD must be replaced.
 - MOV % loss & disconnection indication (Green / Orange / Red)
 - Red Service indicator alarm
- Advanced Thermally Stacked MOV(s)
- Rotatable Color View LCD display. Orientation for varying mounting configurations.
 - Single button control interface
 - Optional Remote monitoring cable for secondary alarm display.
 - See LCD Operational Manual in Annex 1 for more details.
- TOV Surge Discriminator Detection & Counter
 - Up to 10,000 events "Surge TOV, Phase loss, outage"
 - Resettable Counter
- Phase loss indication
- N-G Voltage Detection > 20V detected
- Audible alarm w/ silence option
 - Local audible alarm with a mute option activates upon the operation of any internal disconnecter indicating that the SPD must be replaced.
- Dry Contacts: Remote dry (volt-free) relay contacts rated at ≤ 2 A, 250Vac / 220Vdc, 1 Form C changes state upon any internal disconnecter operating indicating to a customer installed remote indication circuit that the SPD must be replaced.
 - Normally Open / Normally Closed
 - Normally Open = Open during normal operation
 - Normally Closed = Closed during normal operation
- Should any one phase be lost other than electrical anomalies and system issues, the SPD status indicators may change state providing false indication of the SPD end of life. Once all the phases of the electrical system are restored, and if the SPD is not damaged, the status indicators should reset automatically.

Mechanical/Environmental

Operating Temperatures	-40°F (-40°C) to +149°F (+65°C)
Altitude	< 13,000 ft (datasheet value based on C&C calculations)
Surface Temperature	Less than 131°F (55°C)
Humidity	0 - 95% RH
Noise	No Audible Noise
NEMA/UL50 Enclosure	NEMA 1, 4, 4X/ Type 1, 4, 4X
SPD Enclosure Material	Powder Coated Steel, Stainless Steel
Weight	23.4lbs (Approx.)

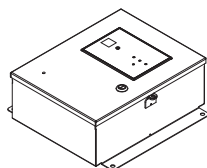
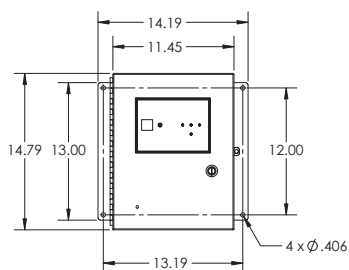
Connections

Remote contact wire size Min-Max	#22-#12 AWG [.08mm ² -4mm ²]
Ground and Neutral Termination Type	Set Screw Lugs
	Min-Max Wire Size: 14 AWG – 2 AWG
	Torque Requirements: 35 in-lb
Phase Wire Size	#10 – #2 AWG

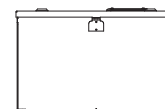
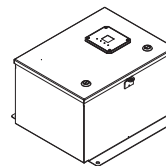
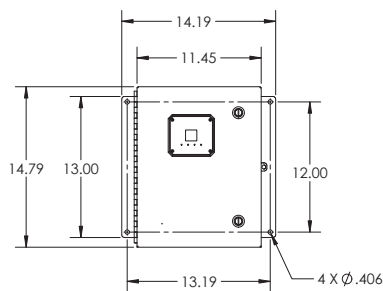


Product Specifications

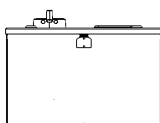
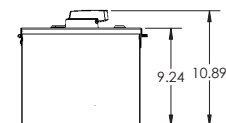
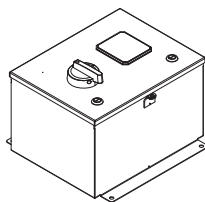
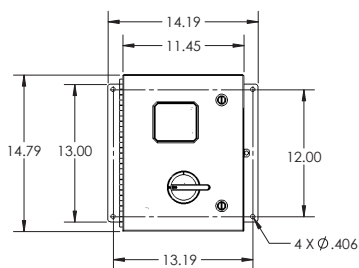
Dimensions



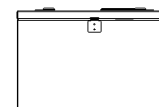
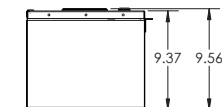
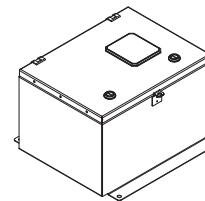
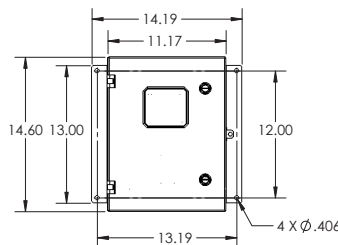
NEMA 1 without Disconnect



NEMA 4 without Disconnect



NEMA 4 with Disconnect



NEMA 4X with Disconnect

SPD Part Number Configuration

*For a full list of orderable products please visit this [resource](#)

RGWMSP

120Y

15

T1

4D

ReliaGear Surge Protection Devices

- ReliaGear Wall Mount Only -

Code	RGWMSP	ReliaGear Wall Mount SPD
------	--------	--------------------------

Nominal Voltage	
Code	(Volt RMS)
120S	120/240
120Y	120Y/208
240D	240 Delta
240H	120/240 Delta HL
277Y	277Y/480
347Y	347Y/600
480D	480 Delta
600D	600 Delta

Max Surge Capacity	
Code	(per mode per phase)
08	80kA 160kA
15	150kA 300kA
22**	225kA 450kA
30**	300kA 600kA

**Available in NEMA 4 and 4X enclosures only

Type	
Code	
T1	UL Type 1
T2	UL Type 2

4D	
Code	
1	NEMA 1 without Disconnect
1D	NEMA 1 with Internal Handle Disconnect
4	NEMA 4 without Disconnect
4D	NEMA 4 with External Handle Disconnect
XD	NEMA 4X with Internal Handle Disconnect

Troubleshooting

There are no serviceable parts inside the SPD. Should SPD indicator LEDs change to RED during its lifetime, the SPD unit will have to be replaced. Please see the instruction manual for further replacement instructions.

RGWMSP Series Truth Table

Operational "OK"	Audible Alarm: NONE
	LEDs: Green—ON
	Red Service Light: OFF
	Dry Contact: NORMAL
MOV Disconnect	Audible Alarm: Beeps
	Either Orange or Red. Depends on remaining percent protection
	Red Service Light: Flashes
	Dry Contact: Triggered
Phase Loss	Audible Alarm: Beeps
	LED color remains at current percent protection but will blink to denote which phase is out
	Red Service Light: Flashes
	Dry Contact: Triggered
N-G voltage (> 20V detected) TOV (Temporary Over Voltage) Surge Event	Audible Alarm: Beeps
	LEDs: Green—ON
	Red Service Light: Flashes
	Dry Contact: Triggered
N-G Voltage / TOV Sustained Event	Audible Alarm: Beeps
	LEDs: Green—ON
	Red Service Light: Flashes
	Dry Contact: Triggered during duration of event
MOV Life Remaining	LED(s) Status Per Phase
	Green: OK
	Orange: 50-100%
	Red: <50%

Warranty

The **RGWMSP** series Surge Protection Device carries a standard 5-year limited warranty, with an optional 10-year warranty.

ABB standard Ts&Cs (12/18)

Circuit Breaker, Arc Fault Plug-In - THQL1120PAF2 (Qty 33)

Catalog No.	THQL1120PAF2
Rating	1P 20A, 10 kAIC, PON Arc Fault
Qty	33
Status	Submitted as specified



Representative Image

Catalog No. THQL1120PAF2

Description: 1P 20AMP 10KAIC PON ARC FAULT

Home

No content displayed.

MCCB, 120/240V QL - THQL2180

Catalog No.	THQL2180
Rating	2P 80A, 120/240V, 10 kAIC
Status	Submitted as specified



Representative Image

Catalog No. THQL2180
Description: THQL 2 POLE 120/240V 10K IC 80 AMP
UPC No 783164099218
Home > Circuit Breakers > Miniature Circuit Breakers > Q-Line Miniature Circuit Breakers

THQL 2 POLE 120/240V 10K IC 80 AMP

Descriptors

Category	Q-Line Miniature Circuit Breakers
Product Line	Q-Line (Plug-In)
GO Schedule	R5

Specifications

Trip Style	Non-Interchangeable
Frame Type	Q-Line
Amperage	80
System Voltage	120 Vac 120/240 Vac
Poles	2
Trip Function	LI
Continuous Current Rated	Standard
120 Vac Interrupting Rating	10 KAIC
120/240 Vac Interrupting Rating	10 KAIC
Suitable for Reverse Feed	Yes
Long Time	Fixed
Instantaneous	Fixed
Protective Relays	No
Current Metering	No
Special Markings	HACR
GSA Compliance	Yes

Classifications

UL File #	E11592
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MCCB, 120/240V QL - THQL2140 (Qty 2)

Catalog No.	THQL2140
Rating	2P 40A, 120/240V, 10 kAIC
Qty	2
Status	Submitted as specified



Representative Image

Catalog No. **THQL2140**

Description: THQL 2POLE 120/240V 10K IC 40 AMPS

UPC No 783164012934

Home > Circuit Breakers > Miniature Circuit Breakers > Q-Line Miniature Circuit Breakers

Q line circuit breakers are one-inch wide per pole, compact, thermal-magnetic devices designed for residential and commercial applications. The QL breakers are plug-in versions of the Q Line used for connection to load centers and lighting panels. All Q Line circuit breakers feature Quick-make / Quick-break mechanisms, common trip bars, and easy to spot trip indication to ensure safety and reliability. Q Line breakers are available in 1, 2, and 3 pole versions, can be ordered with auxiliary contact and shunt trip accessories, and can be ordered for use in HID applications.

Descriptors

Category	Q-Line Miniature Circuit Breakers
Product Line	Q-Line (Plug-In)
GO Schedule	R5

Specifications

Trip Style	Non-Interchangeable
Frame Type	Q-Line
Amperage	40
System Voltage	120 Vac 120/240 Vac
Poles	2
Trip Function	LI
Continuous Current Rated	Standard
120 Vac Interrupting Rating	10 KAIC
120/240 Vac Interrupting Rating	10 KAIC
Suitable for Reverse Feed	Yes
Long Time	Fixed
Instantaneous	Fixed
Protective Relays	No
Current Metering	No
Special Markings	HACR
GSA Compliance	Yes

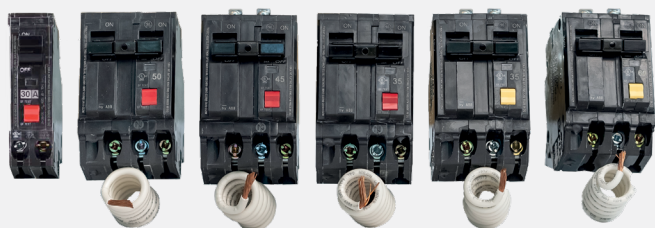
Classifications

UL File #	E11592
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Circuit Breaker, Ground Fault Plug-In - THQL2160GFT2

Catalog No.	THQL2160GFT2
Rating	2P 60A, 120/240V, GFCI
Status	Submitted as specified

Ground Fault Circuit Interrupters



The Ground Fault Circuit Interrupter (GFCI) provides protections against overloads, short circuits, and ground faults. The GFCI breakers detect low levels of electrical current leakage (ground faults), and act quickly to shut off power, with a purpose of preventing serious shock.

The 2023 National Electrical Code (NEC) requires GFCI protection in dwelling units for 125-volt to 250-volt receptacles located:

- Within 6 feet of the top inside edge of the bowl of the sink
- Kitchen receptacles to serve countertops
- Areas with sinks and permanent provisions for food preparation, beverage preparation, or cooking
- Bathrooms
- Within the 6 feet outside edge of bathtubs or shower stalls
- Laundry areas
- Basements
- Garages – including accessory buildings not intended as habitable rooms and limited to storage areas and work areas
- Crawl spaces – at or below grade level
- Outdoors
- Spa areas
- Boathouses
- Indoor damp and wet locations

For exceptions, refer to NEC 2023 article 210.8.

This applies not only to new construction but also to existing homes. When an electrical outlet is replaced in a location that requires ground-fault protection, the new outlet must be GFCI protected, according to the NEC.

Personnel protection

- HACR (Heating, Air Conditioning, and Refrigeration) rated
- UL943 fifth edition
- 5 mA ground fault protection for personnel protection
- Wire range maximum 4 AWG
- Torque requirements:
 - 14 AWG – 12 AWG - 20 lb-in
 - 10 AWG – 30 lb-in
 - 4 AWG – 8 AWG – 40 lb-in

Residential application

	Ampere rating	1-pole, 120 volts AC			2-pole, 120/240 volts AC	
		Long pigtail	Short pigtail	Plug-on neutral	Long pigtail	Plug-on neutral
10,000 AIC	15	THQL1115GFT	THQL1115GFTS	THQL1115PGFT	THQL2115GFT	THQL2115PGFT
	20	THQL1120GFT	THQL1120GFTS	THQL1120PGFT	THQL2120GFT	THQL2120PGFT
	25	THQL1125GFT	—	THQL1125PGFT	THQL2125GFT	THQL2125PGFT
	30	THQL1130GFT	—	THQL1130PGFT	THQL2130GFT	THQL2130PGFT
	35	—	—	—	THQL2135GFT2	—
	40	—	—	—	THQL2140GFT2	—
	45	—	—	—	THQL2145GFT2	—
	50	—	—	—	THQL2150GFT2	—
	60	—	—	—	THQL2160GFT2	—

Residential application

	Ampere rating	1-pole, 120 volts AC			2-pole, 120/240 volts AC	
		Long pigtail	Short pigtail	Plug-on neutral	Long pigtail	Plug-on neutral
22,000 AIC	15	THHQL1115GFT	—	THHQL1115PGFT	THHQL2115GFT	THHQL2115PGFT
	20	THHQL1120GFT	—	THHQL1120PGFT	THHQL2120GFT	THHQL2120PGFT
	25	THHQL1125GFT	—	THHQL1125PGFT	THHQL2125GFT	THHQL2125PGFT
	30	THHQL1130GFT	—	THHQL1130PGFT	THHQL2130GFT	THHQL2130PGFT

Commercial application

	Ampere rating	1-pole, 120 volts AC		2-pole, 120/240 volts AC	
		QB Bolt-on	QC Lug-Lug	QB Bolt-on	QC Lug-Lug
10,000 AIC	15	THQB1115GFT	THQC1115GFT	THQB2115GFT	THQC2115GFT
	20	THQB1120GFT	THQC1120GFT	THQB2120GFT	THQC2120GFT
	25	THQB1125GFT	THQC1125GFT	THQB2125GFT	THQC2125GFT
	30	THQB1130GFT	THQC1130GFT	THQB2130GFT	THQC2130GFT
	35	—	—	THQB2135GFT2	THQC2135GFT2
	40	—	—	THQB2140GFT2	THQC2140GFT2
	45	—	—	THQB2145GFT2	—
	50	—	—	THQB2150GFT2	—
	60	—	—	THQB2160GFT2	—
22,000 AIC	15	THHQB1115GFT	—	THHQB2115GFT	THHQC2115GFT
	20	THHQB1120GFT	—	THHQB2120GFT	THHQC2120GFT
	25	THHQB1125GFT	—	THHQB2125GFT	THHQC2125GFT
	30	THHQB1130GFT	—	THHQB2130GFT	THHQC2130GFT

Equipment protection

- Ground fault
- UL1053 seventh edition
- 30 mA ground fault protection for equipment protection
- Wire range maximum 4 AWG
- Torque requirements:
 - 14 AWG - 12 AWG - 20 lb-in
 - 10 AWG - 30 lb-in
 - 4 AWG - 8 AWG - 40 lb-in

	Ampere rating	1-pole, 120 volts AC			2-pole, 120/240 volts AC		
		QL Plug-on	QB Bolt-on	QC Lug-lug	QL Plug-on	QB Bolt-on	QC Lug-lug
10,000 AIC	15	THQL1115GFEP	THQB1115GFEP	THQC1115GFEP	THQL2115GFEP	THQB2115GFEP	THQC2115GFEP
	20	THQL1120GFEP	THQB1120GFEP	THQC1120GFEP	THQL2120GFEP	THQB2120GFEP	THQC2120GFEP
	30	THQL1130GFEP	THQB1130GFEP	THQC1130GFEP	THQL2130GFEP	THQB2130GFEP	THQC2130GFEP
	35	—	—	—	—	THQB2135GFEP2	—
	40	—	—	—	—	THQB2140GFEP2	—
	45	—	—	—	—	THQB2145GFEP2	—
	50	—	—	—	—	THQB2150GFEP2	—
	60	—	—	—	—	THQB2160GFEP2	—
22,000 AIC	15	THHQL1115GFEP	THHQB1115GFEP	THHQC1115GFEP	THHQL2115GFEP	THHQB2115GFEP	THHQC2115GFEP
	20	THHQL1120GFEP	THHQB1120GFEP	THHQC1120GFEP	THHQL2120GFEP	THHQB2120GFEP	THHQC2120GFEP
	30	THHQL1130GFEP	THHQB1130GFEP	THHQC1130GFEP	THHQL2130GFEP	THHQB2130GFEP	THHQC2130GFEP

ABB Inc.
305 Gregson Drive
Cary, NC 27511

[electrification.us.abb.com/
residential-circuit-breakers](http://electrification.us.abb.com/residential-circuit-breakers)

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MCCB, 120/240V QL - THQL2120

Catalog No.	THQL2120
Rating	2P 20A, 120/240V, 10 kAIC
Status	Submitted as specified



Representative Image

Catalog No. **THQL2120**

Description: THQL 2 POLE 120/240V 10K IC 20 AMP

UPC No 783164012910

Home > Circuit Breakers > Miniature Circuit Breakers > Q-Line Miniature Circuit Breakers

Q line circuit breakers are one-inch wide per pole, compact, thermal-magnetic devices designed for residential and commercial applications. The QL breakers are plug-in versions of the Q Line used for connection to load centers and lighting panels. All Q Line circuit breakers feature Quick-make / Quick-break mechanisms, common trip bars, and easy to spot trip indication to ensure safety and reliability. Q Line breakers are available in 1, 2, and 3 pole versions, can be ordered with auxiliary contact and shunt trip accessories, and can be ordered for use in HID applications.

Descriptors

Category	Q-Line Miniature Circuit Breakers
Product Line	Q-Line (Plug-In)
GO Schedule	R5

Specifications

Trip Style	Non-Interchangeable
Frame Type	Q-Line
Amperage	20
System Voltage	120 Vac 120/240 Vac
Poles	2
Trip Function	LI
Continuous Current Rated	Standard
120 Vac Interrupting Rating	10 KAIC
120/240 Vac Interrupting Rating	10 KAIC
Suitable for Reverse Feed	Yes
Long Time	Fixed
Instantaneous	Fixed
Protective Relays	No
Current Metering	No
Special Markings	HACR
GSA Compliance	Yes

Classifications

UL File #	E11592
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MCCB, 120/240V QL - THQL2130 (Qty 2)

Catalog No.	THQL2130
Rating	2P 30A, 120/240V, 10 kAIC
Qty	2
Status	Submitted as specified



Representative Image

Catalog No. **THQL2130**

Description: THQL 2 POLE 120/240V 10K IC 30 AMP

UPC No 783164012927

Home > Circuit Breakers > Miniature Circuit Breakers > Q-Line Miniature Circuit Breakers

Q line circuit breakers are one-inch wide per pole, compact, thermal-magnetic devices designed for residential and commercial applications. The QL breakers are plug-in versions of the Q Line used for connection to load centers and lighting panels. All Q Line circuit breakers feature Quick-make / Quick-break mechanisms, common trip bars, and easy to spot trip indication to ensure safety and reliability. Q Line breakers are available in 1, 2, and 3 pole versions, can be ordered with auxiliary contact and shunt trip accessories, and can be ordered for use in HID applications.

Descriptors

Category	Q-Line Miniature Circuit Breakers
Product Line	Q-Line (Plug-In)
GO Schedule	R5

Specifications

Trip Style	Non-Interchangeable
Frame Type	Q-Line
Amperage	30
System Voltage	120 Vac 120/240 Vac
Poles	2
Trip Function	LI
Continuous Current Rated	Standard
120 Vac Interrupting Rating	10 KAIC
120/240 Vac Interrupting Rating	10 KAIC
Suitable for Reverse Feed	Yes
Long Time	Fixed
Instantaneous	Fixed
Protective Relays	No
Current Metering	No
Special Markings	HACR
GSA Compliance	Yes

Classifications

UL File #	E11592
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